Coding exercise 1: descriptive statistics, correlation and regression

Individual · 5 % of the final grade · International Marketing Analytics · WU Vienna · WT 2026/27

In brief

Work	individual
Weight	5 % of the final grade (20 points)
Template	`exercise1_template.qmd` next to this brief: save a copy as `exercise1_lastname.qmd` and fill it in
Submit	`exercise1_lastname.qmd` **and** the rendered `.html` on Canvas
Deadline	see Canvas
Data	`data/legacy/chocolate_dataset.csv` in the course repository: 68 weeks of real scanner data, unit sales of brand 1 and prices, features and displays of four chocolate brands
Tools	polars, great_tables, scipy, plotnine and statsmodels

You apply the five building blocks of Session 1 to the chocolate data: **descriptive statistics, correlation, scatterplots, linear regression and log-log regression**. Each part below starts with a **code element**: a short, working example of the functions you need. The examples use other variables than the task, so adapt them; do not copy them unchanged. Your templates from class: `sessions/01-foundations/stats_refresher.qmd` (Step 6) and `sessions/01-foundations/regression_chocolate.qmd`.

Set-up

Start from the template `exercise1_template.qmd`: it already contains the set-up below, one section per part with empty code cells, and the prompt-log table.

Install once (skip if you did it in class): pick **Python 3.14 (.venv)** at the top right of Positron and run this line in a cell or the Console. In your submitted file, keep it in a cell with `#| eval: false`, so that Quarto does not run it when rendering.

```
%pip install polars plotnine great-tables statsmodels pyarrow jupyter
```

Load at the top of your file, in the first code cell, and read the data:

```
import numpy as np                          # np.log in model formulas
import polars as pl                         # data and descriptive statistics
```

```
import statsmodels.formula.api as smf    # regression
from great_tables import GT              # tables
from plotnine import aes, geom_point, geom_smooth, ggplot, labs   # charts
from scipy import stats                  # p-value of a correlation

choc = pl.read_csv("../data/legacy/chocolate_dataset.csv")      # path from the exercises/
```

Save your file in the `exercises/` folder of the course repository, next to this brief, so the path above works; in the Positron Console, run from the project folder, the path is `data/legacy/chocolate_dataset.csv`. scipy and numpy come with statsmodels. Install `great-tables`, load `great_tables`. Error *No module named …*? See Step 0 of `regression_chocolate.qmd`.

Use these five variables in Parts 1 and 2: `sales`, `price1`, `feature1`, `display1`, `temp`.

Part 1: descriptive statistics

Code element. `describe()` gives count, mean, standard deviation, min, quartiles (the 50 % row is the median) and max; `GT` turns the result into a clean table.

```
example = ["sales", "price2"]
summary = (choc.select(example).describe()
           .filter(pl.col("statistic").is_in(["mean", "std", "min", "50%", "max"])))
GT(summary).fmt_number(columns=example, decimals=2).tab_header(title="Example")
```

Task 1. A great_tables table with the mean, standard deviation, minimum, median and maximum of the five variables. In two sentences: compare mean and median of `sales` (what does the difference tell you?), and say what the means of `feature1` and `display1` (shares of stores, 0 to 1) tell you about how often brand 1 used them.

Part 2: correlation

Code element. `corr()` gives the correlation matrix of all selected columns; scipy's `pearsonr` gives r and its p-value for one pair.

```
choc.select("sales", "price2", "price3").corr()
r, p = stats.pearsonr(choc["sales"], choc["price2"])
```

Task 2. The correlation matrix of the five variables, and r with p-value for `sales` and `display1`. Which variable moves most strongly with sales, and in which direction? Is the correlation of sales and display significant at 5 %? Look at the correlation of `price1` and `feature1`: why could it matter for the regressions in Parts 4 and 5?

Part 3: scatterplots

Code element. A scatterplot with the least-squares line; `with_columns` adds log variables.

```r
(ggplot(choc, aes("price2", "sales"))
 + geom_point()
 + geom_smooth(method="lm")
 + labs(x="Price of brand 2 (EUR)", y="Weekly sales of brand 1"))

logs = choc.with_columns(pl.col("sales").log().alias("log_sales"))
```

Task 3. Two scatterplots with the least-squares line, clearly labelled: `sales` against `price1`, and log sales against log `price1`. Which of the two is closer to a straight line, and what does that mean for the choice of model?

Part 4: linear regression

Code element. statsmodels' formula API takes a pandas data frame: `.to_pandas()` converts the polars frame.

```r
df = choc.to_pandas()
example = smf.ols("sales ~ temp", data=df).fit()
print(example.summary())
example.params["temp"], example.pvalues["temp"], example.rsquared
```

Task 4. Estimate `sales ~ price1 + feature1`. Interpret the price coefficient for a price change of **10 cents** and the `feature1` coefficient (a change from 0 to 1 means from no store to all stores). Which coefficients are significant at 5 %? Report R^2. Use your answer to Task 2 to explain the `feature1` result.

Part 5: log-log regression

Code element. `np.log` inside the formula transforms the variables; in a log-log model the coefficient is an elasticity. A dummy in a log model shifts sales by exp(b) − 1.

```r
example = smf.ols("np.log(sales) ~ np.log(price2) + easter", data=df).fit()
eps = example.params["np.log(price2)"]  # an elasticity
np.exp(example.params["easter"]) - 1    # dummy effect (one Easter week only)
```

Task 5. Estimate

`np.log(sales) ~ np.log(price1) + np.log(price2) + np.log(price3) + np.log(price4) + feature1 + display1 + temp + december`.

a. Interpret the own-price elasticity, one competitor price and `december` (exp(b) − 1). Which coefficients are significant at 5 %?

b. The coefficient of `display1` is large and significant. Check in the data how many weeks had a display (`(choc["display1"] > 0).sum()`): how much do you trust it?

c. With a marginal cost of 0.90, compute the price that the own-price elasticity suggests ($p^* = \text{cost} \times \varepsilon / (1 + \varepsilon)$). In at most 100 words, recommend a regular price to the brand manager and give two caveats.

Rules

- Every number in your text comes from code (inline code such as `` `{python} f"{eps:.2f}"` ``).
- The file renders from top to bottom without errors; charts and tables have clear labels.
- AI assistants (Posit Assistant with Copilot) are allowed; you own every number. End the file with a **prompt log**: the prompts you used, and what you kept or changed.
- Individual work: discuss ideas, but write and check your own code. Questions go to the Q&A forum on Canvas.

Grading (20 points)

Part	Points
Code runs from top to bottom; charts and tables clean and labelled	4
1 Descriptive statistics: table and interpretation	2
2 Correlation: matrix, p-value and interpretation	3
3 Scatterplots: two charts and the choice of model	2
4 Linear regression: interpretation (units, significance, R^2)	3
5 Log-log regression: elasticities, dummy, display check, recommendation	4
Prompt log	2
Total	**20**